

Complete Production Deployment Guide

Oracle Database 21c XE + Oracle APEX 26.1 + ORDS + Nginx + HTTPS on Hostinger VPS KVM 2 (Ubuntu + Docker)

Overview

This document describes the complete deployment process for running a production-style Oracle APEX environment on a Hostinger VPS KVM 2 server using:

- Oracle Database 21c XE
- Oracle APEX 26.1
- Oracle REST Data Services (ORDS) 26.1
- Docker + Docker Compose
- Nginx reverse proxy
- Let's Encrypt SSL
- Ubuntu 22.04 LTS

This guide includes:

- working configurations
- important fixes
- issues encountered
- solutions implemented
- final production architecture
- reverse proxy fixes
- HTTPS fixes
- ORDS CORS fixes

The deployment was validated successfully.

Server Specifications

VPS

Hostinger VPS KVM 2

- 2 vCPU
- 8 GB RAM
- 100 GB SSD

Recommended OS:

- Ubuntu 22.04 LTS
-

Final Architecture

```
Internet
  ↓
Nginx HTTPS :443
  ↓
ORDS localhost :8080
  ↓
Oracle XE localhost :1521
  ↓
Oracle APEX
```

Security design:

- Oracle listener not publicly exposed
 - ORDS not publicly exposed directly
 - Nginx acts as reverse proxy
 - HTTPS termination at Nginx
-

STEP 1 — Initial Ubuntu Server Preparation

Update Server

```
sudo apt update && sudo apt upgrade -y
sudo reboot
```

Set Hostname

```
sudo hostnamectl set-hostname apex-server
```

Edit:

```
sudo nano /etc/hosts
```

Add:

```
127.0.0.1 localhost
127.0.1.1 apex-server
```

Create Swap Memory

Oracle benefits greatly from swap.

Create 8GB Swap

```
sudo fallocate -l 8G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
```

Persist swap:

```
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
```

Verify:

```
free -h
```

Install Base Utilities

```
sudo apt install -y
curl
wget
git
nano
unzip
htop
```

```
ca-certificates
gnupg
lsb-release
software-properties-common
tree
```

Configure Firewall

```
sudo apt install ufw -y
sudo ufw allow OpenSSH
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw enable
```

Verify:

```
sudo ufw status
```

Set Timezone

```
sudo timedatectl set-timezone Asia/Kolkata
```

Create Project Directories

```
sudo mkdir -p /opt/apex-docker
sudo chown -R $USER:$USER /opt/apex-docker
```

```
mkdir -p /opt/apex-docker/{oracle,ords,nginx,backups}
```

Kernel Tuning

Edit:

```
sudo nano /etc/sysctl.conf
```

Add:

```
fs.file-max = 6815744
kernel.sem = 250 32000 100 128
kernel.shmmax = 4294967295
kernel.shmall = 1073741824
net.ipv4.ip_local_port_range = 9000 65500
fs.aio-max-nr = 1048576
vm.swappiness=10
```

Apply:

```
sudo sysctl -p
```

Disable Transparent Huge Pages

Create service:

```
sudo nano /etc/systemd/system/disable-thp.service
```

```
[Unit]
Description=Disable Transparent Huge Pages
After=network.target

[Service]
Type=simple
ExecStart=/bin/sh -c "echo never > /sys/kernel/mm/transparent_hugepage/enabled"
ExecStart=/bin/sh -c "echo never > /sys/kernel/mm/transparent_hugepage/defrag"

[Install]
WantedBy=multi-user.target
```

Enable:

```
sudo systemctl daemon-reload
sudo systemctl enable disable-thp
sudo systemctl start disable-thp
```

Verify:

```
cat /sys/kernel/mm/transparent_hugepage/enabled
```

Expected:

```
[never]
```

File Limits

Edit:

```
sudo nano /etc/security/limits.conf
```

Add:

```
* soft nfile 65535
* hard nfile 65535
* soft nproc 65535
* hard nproc 65535
```

systemd Limits

Edit:

```
sudo nano /etc/systemd/system.conf
```

Add:

```
DefaultLimitNOFILE=65535
DefaultLimitNPROC=65535
```

Edit:

```
sudo nano /etc/systemd/user.conf
```

Add same entries.

Reload:

```
sudo systemctl daemon-reexec
sudo systemctl daemon-reload
```

Enable PAM Limits

Edit:

```
sudo nano /etc/pam.d/common-session
```

Add:

```
session required pam_limits.so
```

Also edit:

```
sudo nano /etc/pam.d/common-session-noninteractive
```

Add:

```
session required pam_limits.so
```

STEP 2 — Install Docker

Remove Old Docker Packages

```
sudo apt remove -y docker docker-engine docker.io containerd runc
```

Add Official Docker Repository

```
sudo install -m 0755 -d /etc/apt/keyrings
```

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg |  
sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
```

```
sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

```
echo  
"deb [arch=$(dpkg --print-architecture)  
signed-by=/etc/apt/keyrings/docker.gpg]  
https://download.docker.com/linux/ubuntu  
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" |  
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

```
sudo apt update
```

Install Docker CE

```
sudo apt install -y  
docker-ce  
docker-ce-cli  
containerd.io  
docker-buildx-plugin  
docker-compose-plugin
```

Enable:


```
sudo systemctl enable docker
sudo systemctl start docker
```

Verify:

```
docker --version
docker compose version
```

Docker Daemon Configuration

Create:

```
sudo nano /etc/docker/daemon.json
```

```
{
  "log-driver": "json-file",
  "log-opts": {
    "max-size": "100m",
    "max-file": "5"
  },
  "storage-driver": "overlay2",
  "live-restore": true,
  "default-ulimits": {
    "nofile": {
      "Name": "nofile",
      "Hard": 1048576,
      "Soft": 1048576
    }
  }
}
```

Restart Docker:

```
sudo systemctl restart docker
```

STEP 3 — Oracle Database 21c XE Deployment

Oracle Registry Login

```
docker login container-registry.oracle.com
```

Accept Oracle license in browser first.

Pull Oracle XE Image

```
docker pull container-registry.oracle.com/database/express:21.3.0-xe
```

Final compose.yml

Location:

```
/opt/apex-docker/compose.yml
```

```
services:

  oracle-db:
    image: container-registry.oracle.com/database/express:21.3.0-xe

    container_name: oracle-xe
    hostname: oracle-xe

    restart: unless-stopped

    networks:
      - apex-net

    ports:
      - "127.0.0.1:1521:1521"

    environment:
      ORACLE_PWD: "DaljsAdmin26"
      ORACLE_CHARACTERSET: "AL32UTF8"
```

```

volumes:
  - /opt/apex-docker/oracle/data:/opt/oracle/oradata

shm_size: "2gb"

ulimits:
  nofile:
    soft: 65535
    hard: 65535

healthcheck:
  test: >
    /bin/bash -c "echo 'SELECT 1 FROM dual;' |
    sqlplus -s sys/DaljsAdmin26@localhost/XEPDB1 as sysdba |
    grep -q 1"

  interval: 30s
  timeout: 10s
  retries: 10
  start_period: 10m

ords:
  image: container-registry.oracle.com/database/ords:latest

  container_name: ords
  hostname: ords

  restart: unless-stopped

  depends_on:
    oracle-db:
      condition: service_healthy

  networks:
    - apex-net

  ports:
    - "127.0.0.1:8080:8080"

  environment:
    ORDS_CONFIG: /etc/ords/config

  volumes:
    - /opt/apex-docker/ords/config:/etc/ords/config
    - /opt/apex-docker/ords/images:/opt/oracle/ords/images
    - /opt/apex-docker/ords/logs:/tmp/ords

  command: >

```

```
--config /etc/ords/config serve
```

```
networks:  
  apex-net:
```

IMPORTANT Oracle Permission Fix

The following issue occurred:

```
Cannot create directory "/opt/oracle/oradata/XE"
```

Cause:

Docker bind mount permissions.

Fix:

```
sudo chown -R 54321:54321 /opt/apex-docker/oracle/data  
sudo chmod -R 775 /opt/apex-docker/oracle/data
```

Clean failed initialization:

```
sudo rm -rf /opt/apex-docker/oracle/data/*
```

Start Oracle

```
cd /opt/apex-docker  
docker compose up -d
```

Monitor logs:

```
docker logs -f oracle-xe
```

Wait for:

DATABASE IS READY TO USE!

STEP 4 — Install Oracle APEX

Download APEX

Download from Oracle:

- Oracle APEX 26.1

Extract:

```
unzip apex-latest.zip
```

Copy Into Container

```
docker cp /opt/apex-docker/apex oracle-xe:/opt/oracle/
```

Install APEX

Enter container:

```
docker exec -it oracle-xe bash
```

Go to APEX directory:

```
cd /opt/oracle/apex
```

Connect SQLPlus:

```
sqlplus sys/DaljsAdmin26@XEPDB1 as sysdba
```

IMPORTANT:

Initial install attempt failed because apexins.sql was executed using absolute path.

Correct method:

- cd into APEX folder first
- then run script locally

Correct install:

```
@apexins.sql SYSaux SYSaux TEMP /i/
```

Using SYSaux tablespace is acceptable for single-VPS deployment.

Configure APEX

Set admin password:

```
@apxchpwd.sql
```

Unlock APEX_PUBLIC_USER:

```
ALTER USER APEX_PUBLIC_USER ACCOUNT UNLOCK;  
ALTER USER APEX_PUBLIC_USER IDENTIFIED BY "DaljsAdmin26#";
```

Configure authentication:

```
BEGIN  
  APEX_INSTANCE_ADMIN.SET_PARAMETER(  
    'APEX_BUILDER_AUTHENTICATION',  
    'APEX'  
  );  
  COMMIT;  
END;  
/
```

Verify version:

```
SELECT version_no FROM apex_release;
```

Copy APEX Images

```
mkdir -p /opt/apex-docker/ords/images
```

```
cp -r /opt/apex-docker/apex/images/*  
/opt/apex-docker/ords/images/
```

STEP 5 — Install ORDS

ORDS was first installed interactively to simplify debugging.

ORDS Permission Fix

ORDS initially failed because:

```
/etc/ords/config is not writable
```

Fix:

```
sudo chown -R 54321:54321 /opt/apex-docker/ords  
sudo chmod -R 775 /opt/apex-docker/ords
```

Interactive ORDS Install

Run:

```
docker run -it --rm  
--name ords-install  
--network apex-docker_apex-net  
-p 8080:8080  
-v /opt/apex-docker/ords/config:/etc/ords/config  
-v /opt/apex-docker/ords/images:/opt/oracle/ords/images
```

```
--entrypoint /bin/bash
container-registry.oracle.com/database/ords:latest
```

Inside container:

```
ords --config /etc/ords/config install
```

Use:

- Host: oracle-db
- Port: 1521
- Service: XEPDB1
- Gateway mode: proxied
- APEX images path: /opt/oracle/ords/images

IMPORTANT ORDS Fix

Initial Docker Compose startup failed with:

```
Unknown command: ords ords
```

Cause:

ORDS image already has ords as entrypoint.

Wrong:

```
command: >
ords --config /etc/ords/config serve
```

Correct:

```
command: >
--config /etc/ords/config serve
```

STEP 6 — Install Nginx

Install:

```
sudo apt install nginx -y
```

Enable:

```
sudo systemctl enable nginx  
sudo systemctl start nginx
```

Domain DNS

DNS configuration:

Type	Host	Value
A	@	VPS IP
CNAME	www	domain.com

Final Nginx Config

Location:

```
/etc/nginx/sites-available/apex
```

```
server {  
  
    listen 80;  
    server_name daljssoft.me www.daljssoft.me;  
  
    return 301 https://$host$request_uri;  
}  
  
server {  
  
    listen 443 ssl http2;
```

```

server_name daljssoft.me www.daljssoft.me;

client_max_body_size 200M;

ssl_certificate /etc/letsencrypt/live/daljssoft.me/fullchain.pem;
ssl_certificate_key /etc/letsencrypt/live/daljssoft.me/privkey.pem;

ssl_session_timeout 1d;
ssl_session_cache shared:MozSSL:10m;

ssl_protocols TLSv1.2 TLSv1.3;

add_header X-Frame-Options SAMEORIGIN;
add_header X-Content-Type-Options nosniff;
add_header X-XSS-Protection "1; mode=block";

location / {

    proxy_pass http://127.0.0.1:8080;

    proxy_http_version 1.1;

    proxy_set_header Host $host;
    proxy_set_header X-Forwarded-Host $host;
    proxy_set_header X-Forwarded-Server $host;
    proxy_set_header X-Forwarded-Proto https;
    proxy_set_header X-Forwarded-Port 443;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;

    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection "upgrade";

    proxy_redirect off;

    proxy_buffering off;

    proxy_read_timeout 3600;
    proxy_send_timeout 3600;
}
}

```

Enable site:

```
sudo ln -s /etc/nginx/sites-available/apex  
/etc/nginx/sites-enabled/
```

Remove default:

```
sudo rm -f /etc/nginx/sites-enabled/default
```

Test:

```
sudo nginx -t
```

Restart:

```
sudo systemctl restart nginx
```

STEP 7 — HTTPS SSL

Install Certbot:

```
sudo apt install certbot python3-certbot-nginx -y
```

Generate SSL:

```
sudo certbot --nginx
```

Enable:

- redirect HTTP → HTTPS

Verify renewal:

```
sudo certbot renew --dry-run
```

IMPORTANT ORDS HTTPS FIX

After HTTPS, login failed with:

```
ORDS-13002  
Cross Origin Sharing requests
```

Cause:

ORDS 26.x stricter reverse proxy validation.

FINAL ORDS settings.xml FIX

File:

```
/opt/apex-docker/ords/config/global/settings.xml
```

Final working version:

```
<?xml version="1.0" encoding="UTF-8"?>  
<!DOCTYPE properties SYSTEM "http://java.sun.com/dtd/properties.dtd">  
  
<properties>  
  
    <comment>ORDS Global Settings</comment>  
  
    <entry key="database.api.enabled">true</entry>  
  
    <entry key="standalone.doc.root">/etc/ords/config/global/doc_root</entry>  
  
    <entry key="standalone.http.port">8080</entry>  
  
    <entry key="standalone.static.path">/opt/oracle/ords/images</entry>  
  
    <entry key="standalone.forwardedHeadersEnabled">true</entry>  
  
    <entry key="security.externalSessionTrustedOrigins">  
        https://daljssoft.me,https://www.daljssoft.me  
    </entry>
```

```
</properties>
```

Restart ORDS:

```
cd /opt/apex-docker  
docker compose restart ords
```

This FIXED:

- ORDS-13002
 - HTTPS login failures
 - reverse proxy origin validation
-

Final URLs

APEX:

```
https://daljssoft.me/ords/apex
```

Administration:

```
https://daljssoft.me/ords/apex_admin
```

Final Security State

Public:

- 80
- 443

Internal only:

- 1521
- 8080

ORDS bound to localhost only.

Oracle listener bound to localhost only.

Useful Commands

View Containers

```
docker ps
```

View ORDS Logs

```
docker logs -f ords
```

View Oracle Logs

```
docker logs -f oracle-xe
```

Restart Stack

```
cd /opt/apex-docker  
docker compose restart
```

Stop Stack

```
docker compose down
```

Start Stack

```
docker compose up -d
```

Backup Recommendations

Oracle Data Backup

```
tar -czvf /opt/apex-docker/backups/oracle-backup.tar.gz  
/opt/apex-docker/oracle/data
```

APEX Export Backup

Use APEX export utilities regularly.

Final Notes

This deployment successfully achieved:

- Oracle XE Docker deployment
- Oracle APEX deployment
- ORDS deployment
- Docker persistence
- HTTPS reverse proxy
- ORDS reverse proxy fixes
- SSL secured APEX environment
- production-style architecture

The system is suitable for:

- ERP applications
 - internal business apps
 - student management systems
 - medium production workloads
 - multi-workspace APEX environments
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End of Guide